SHDS: Lecture 14

14. Linear Regression 3

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Introduction

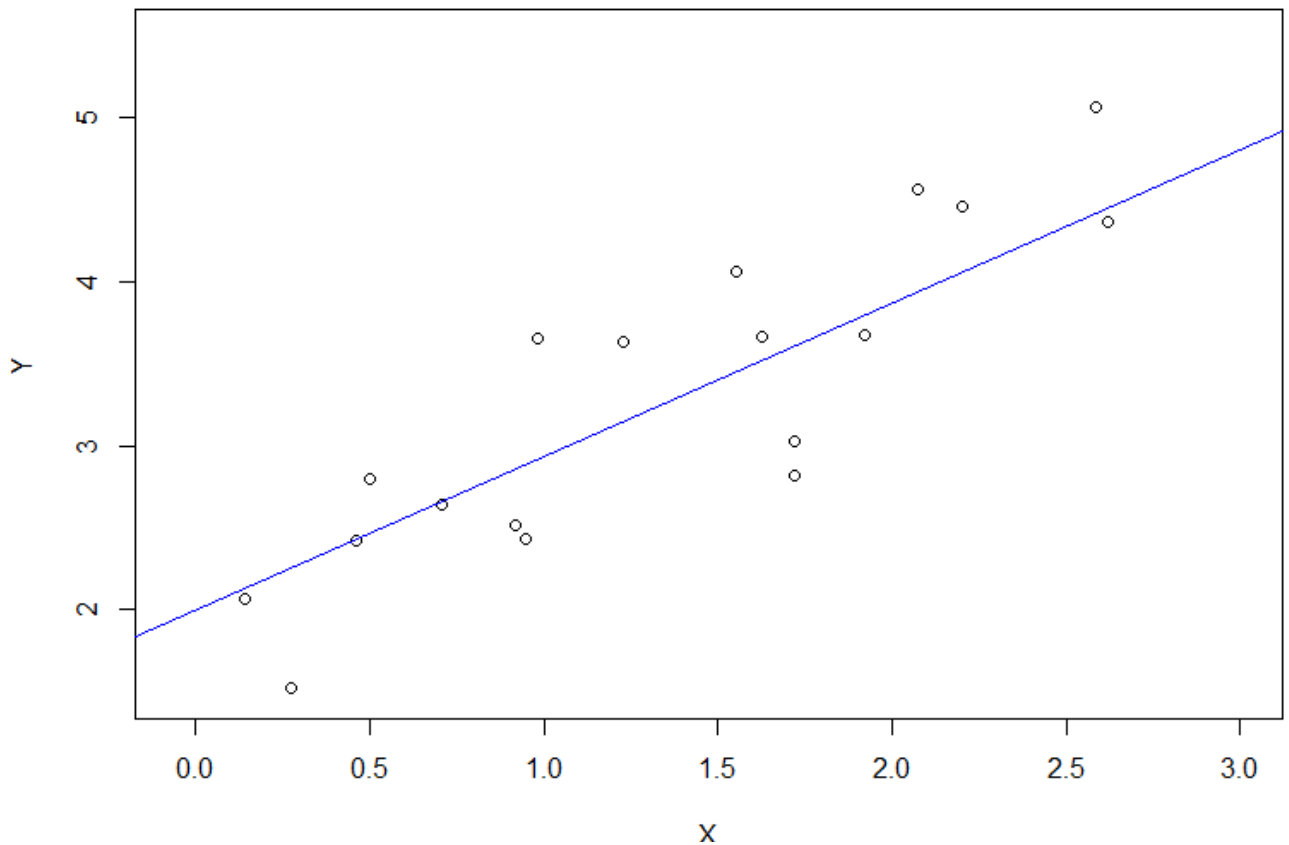
A parametric statistical model is an algebraic description of how one or more outcome variables are influenced by covariates.

Such models are widely used in medical research. Some examples of questions that we can investigate using statistical models include:

- Does birthweight increase with length of pregnancy?
- Does taking drug A reduce inflammation more than taking drug B in patients with arthritis?
- Can we predict the risk of heart disease for our patients?

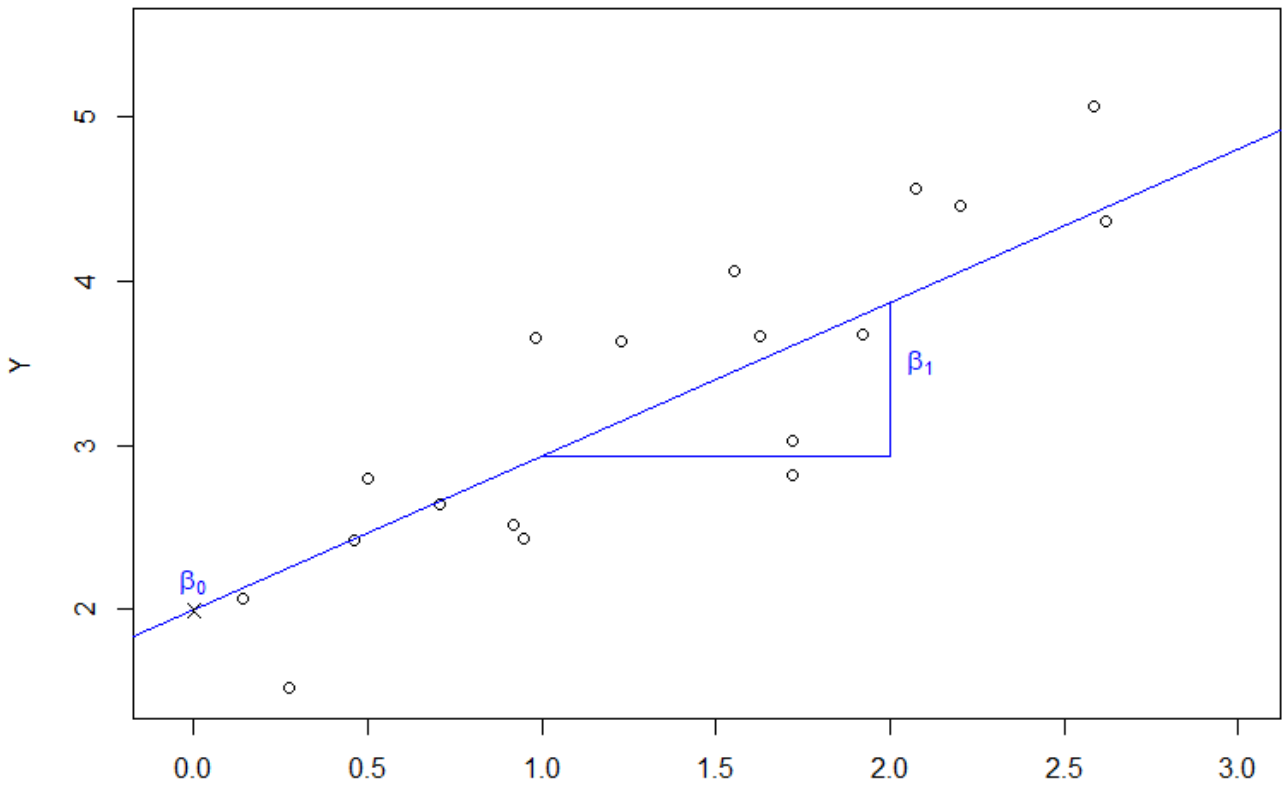
Rationale

- We assume there is a true underlying linear relationship between X and Y and we statistical models to estimate that relationship.



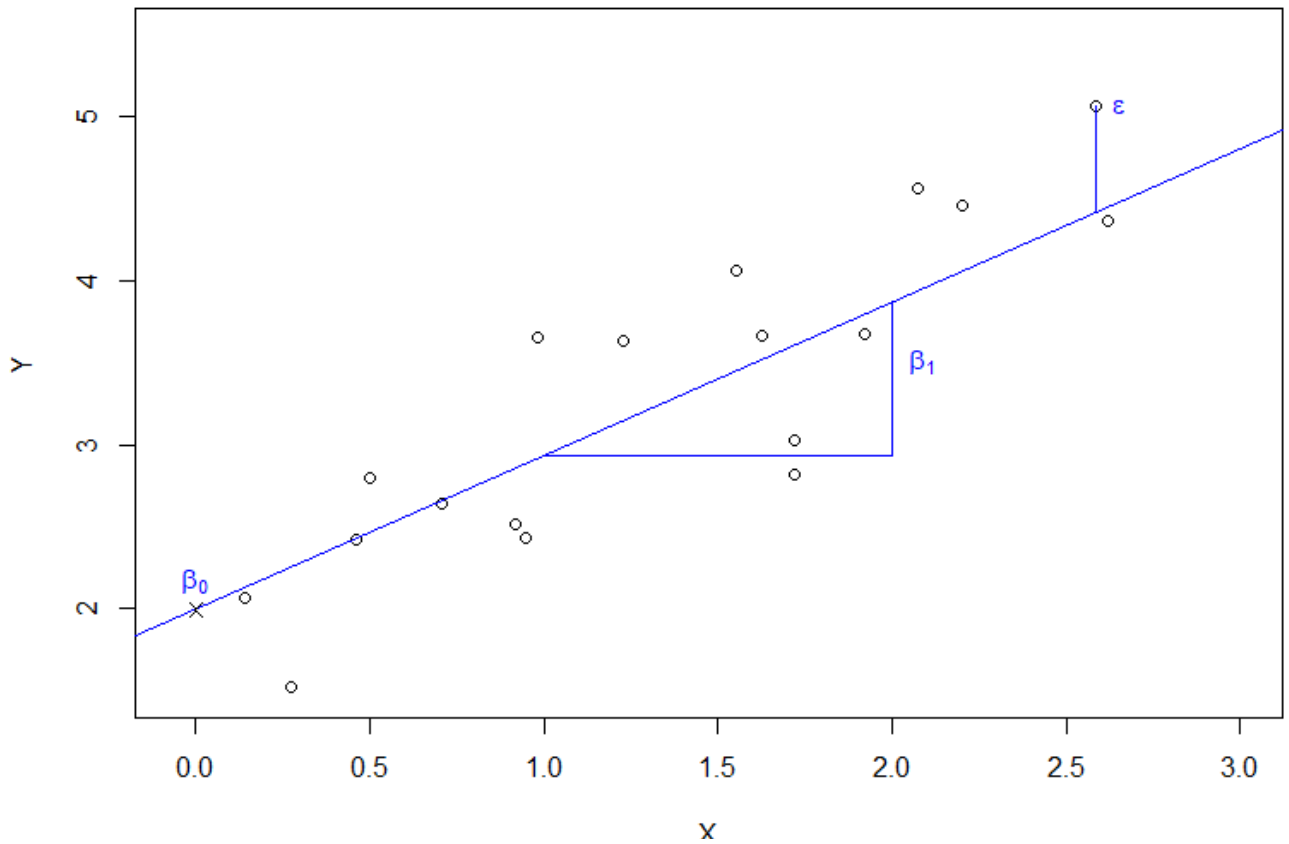
The simple linear regression model

- **Intercept** (β_0): expected value of Y when X is equal to 0
- **Slope** (β_1): expected change in Y for each unit increase in X



The simple linear regression model

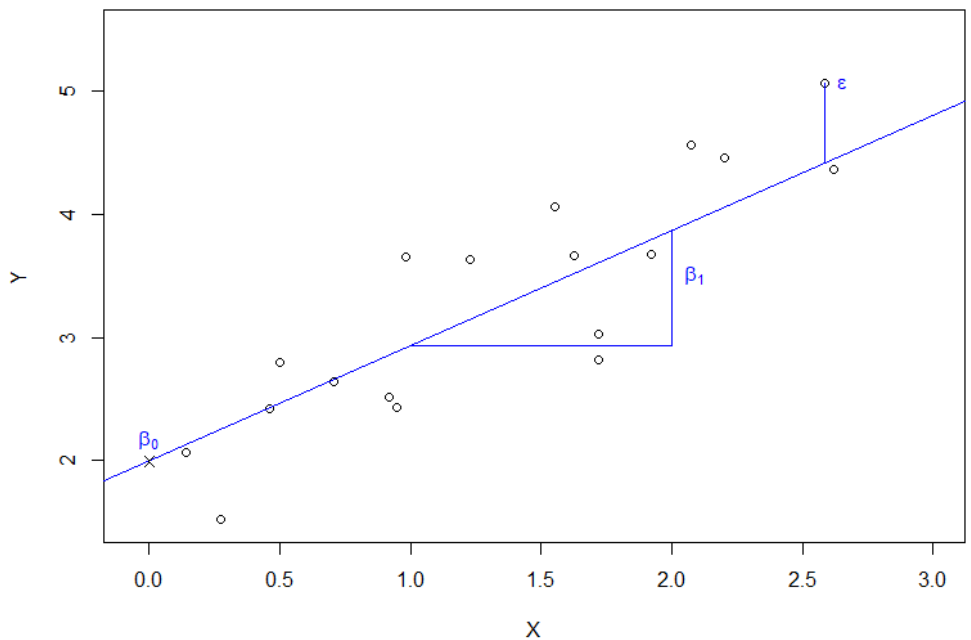
- **Residuals**: difference between observed values and the true regression line (we assume $\epsilon \sim NID(0, \sigma^2)$)



The simple linear regression model

The simple linear regression equation:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i \text{ where } \epsilon_i \sim NID(0, \sigma^2)$$



Assumptions

The simple linear regression equation:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i \text{ where } \epsilon_i \sim NID(0, \sigma^2)$$

Assumptions:

- Linearity
- Normality
- Homoscedasticity
- Independence

Example: data

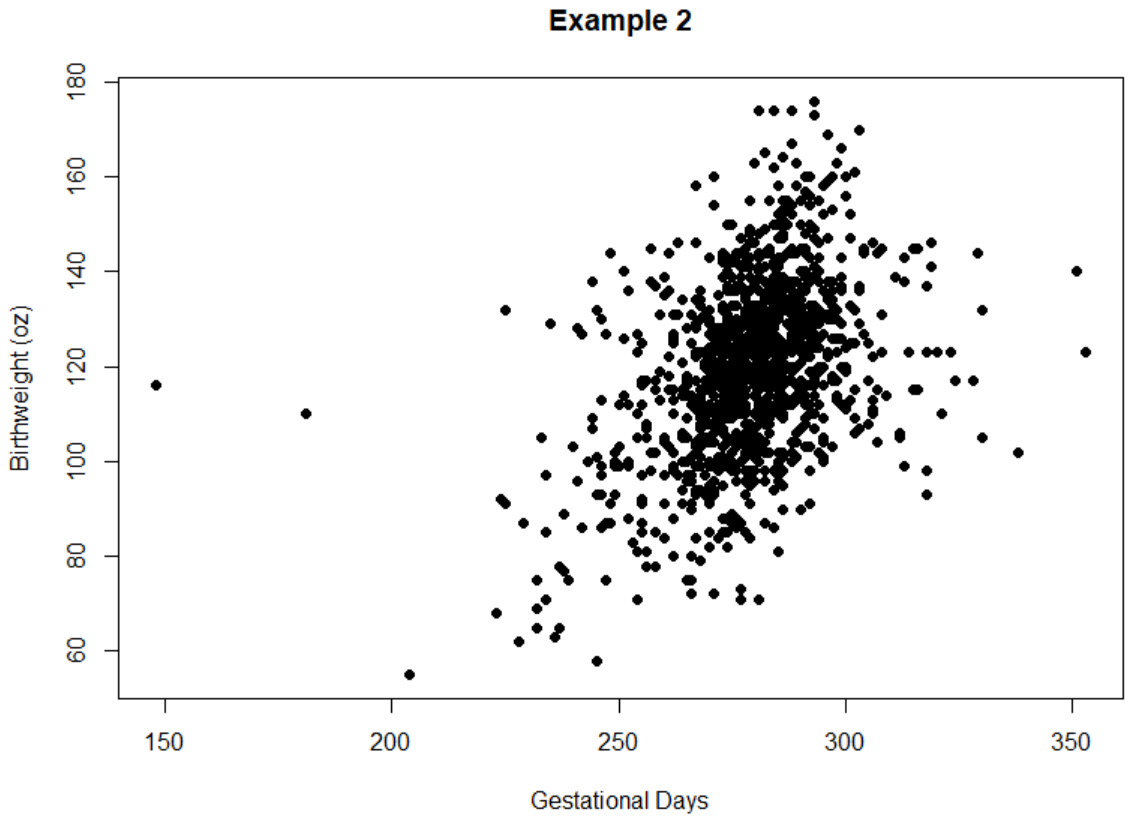
In our second example, we use data on babies and their mothers. This data contains a random sample of 1,174 mothers and their newborn babies. Variables in the data include:

- Birth weight (oz)
- Gestational days
- Maternal age
- Maternal height
- Maternal pregnancy weight
- Whether or not the mother smokes

We want to investigate the association between birth weight and length of pregnancy.

Example: data

- Birth weight ranges from 55 to 176oz, with a median of 120oz
- Gestational days ranges from 148 to 353, with a median of 230



Example: R output

```
> bw_model<-lm(baby_data$Birth.Weight~baby_data$Gestational.Days)
> summary(bw_model)

Call:
lm(formula = baby_data$Birth.Weight ~ baby_data$Gestational.Days)

Residuals:
    Min      1Q  Median      3Q     Max
-49.348 -11.065   0.218  10.101  57.704

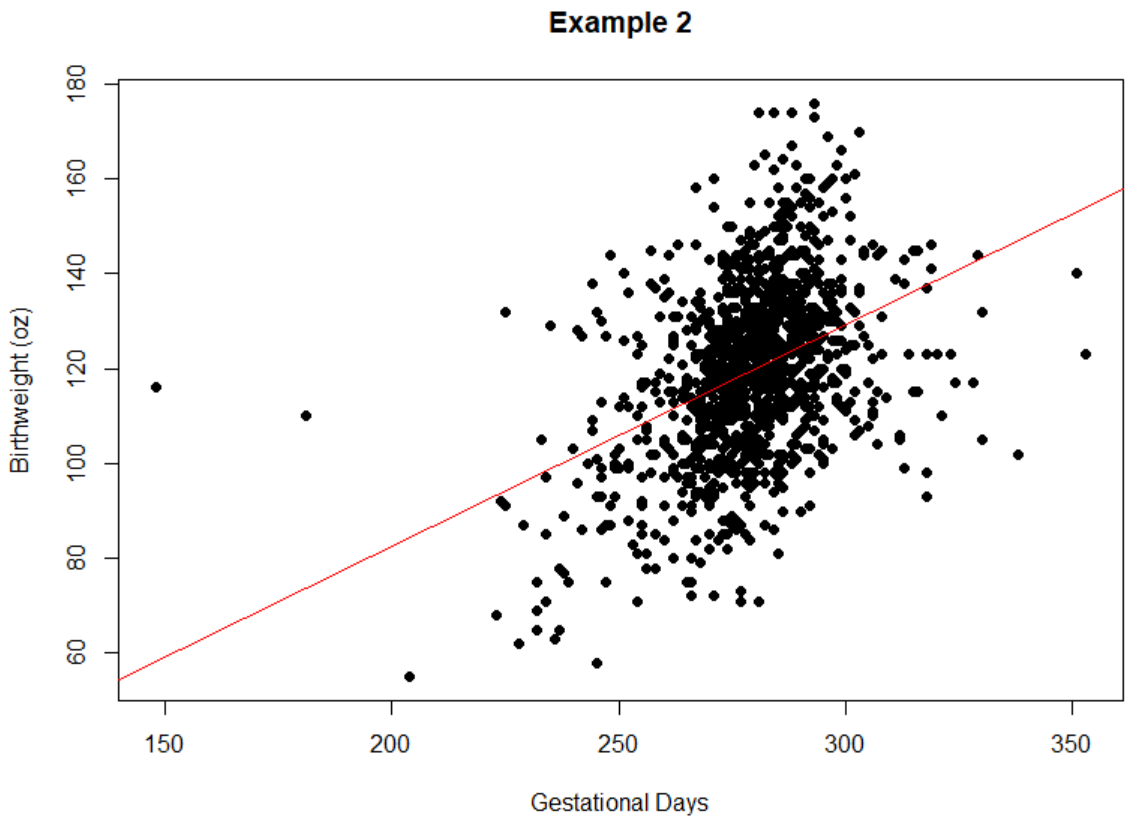
Coefficients:
                           Estimate Std. Error t value Pr(>|t|)
(Intercept)                -10.75414    8.53693   -1.26    0.208
baby_data$Gestational.Days   0.46656    0.03054   15.28   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 16.74 on 1172 degrees of freedom
Multiple R-squared:  0.1661,    Adjusted R-squared:  0.1654
F-statistic: 233.4 on 1 and 1172 DF,  p-value: < 2.2e-16
```

- $\hat{\beta}_0 = -10.75$
- $\hat{\beta}_1 = 0.47$

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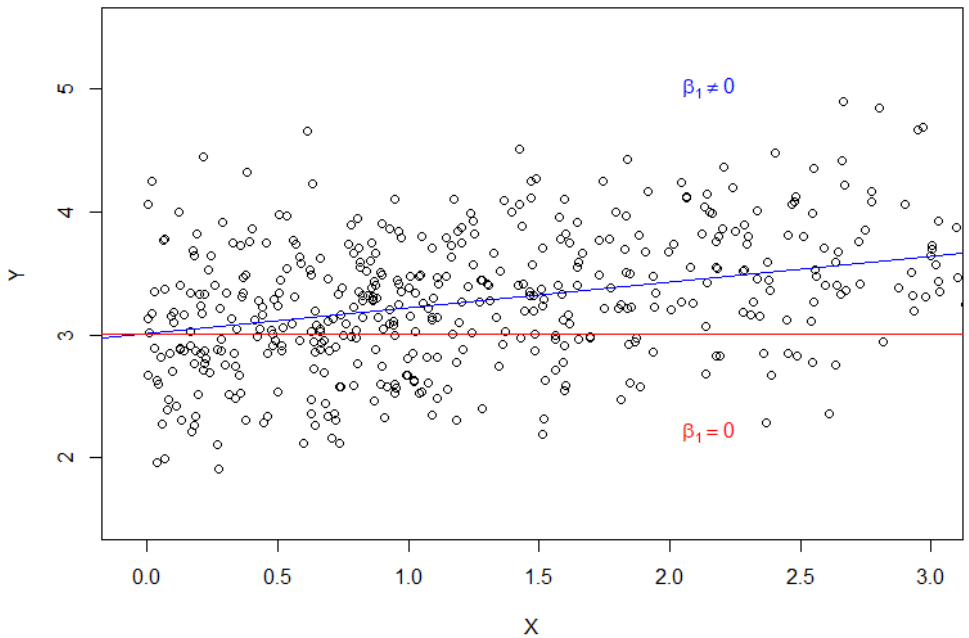


Example: interpreting the parameters

- $\hat{\beta}_0 = -10.75$
- The expected birthweight of a child born after 0 gestational days is -10.75oz
- $\hat{\beta}_1 = 0.47$
- The mean birthweight of a baby is estimated to increase by 0.47oz for each daily increase in the gestational period.
- $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_2 X$
- The expected birthweight for a child born after 200 days is $-10.75 + 0.47 \times 200 = 83.25$

Inference

- Based on the results from our linear regression, what can we infer about the model parameters?
- Most commonly interested in whether $\beta_1 = 0$



- Two approaches: (1) Hypothesis tests and (2) Confidence intervals

Inference

Hypothesis tests

- $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0$
- Test statistic: $t = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)}$ where $SE(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{ns_x^2}}$
- Distribution of the test statistic: $t \sim t_{n-2}$

Confidence intervals (CI)

- 95% CI: $\hat{\beta}_1 \pm t_{\frac{0.05}{2},(n-2)} \times SE(\hat{\beta}_1)$
- 95% CI: $\hat{\beta}_1 \pm 1.96 \times SE(\hat{\beta}_1)$ (for large n)

Example: R output

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- Can reject the null hypothesis that $\beta_1 = 0$
- CIs can be estimated using confint()

Assumptions

The simple linear regression equation:

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- Example: if we carry out a study in which the blood pressure of 100 people are each measured twice, then treat the 200 measurements as independent, this is a clear violation.
- Example: In some medical specialities, such as ophthalmology, otorhinolaryngology, and internal medicine, both eyes of the same patient are often included in analyses, which violates independence

Remaining three assumptions

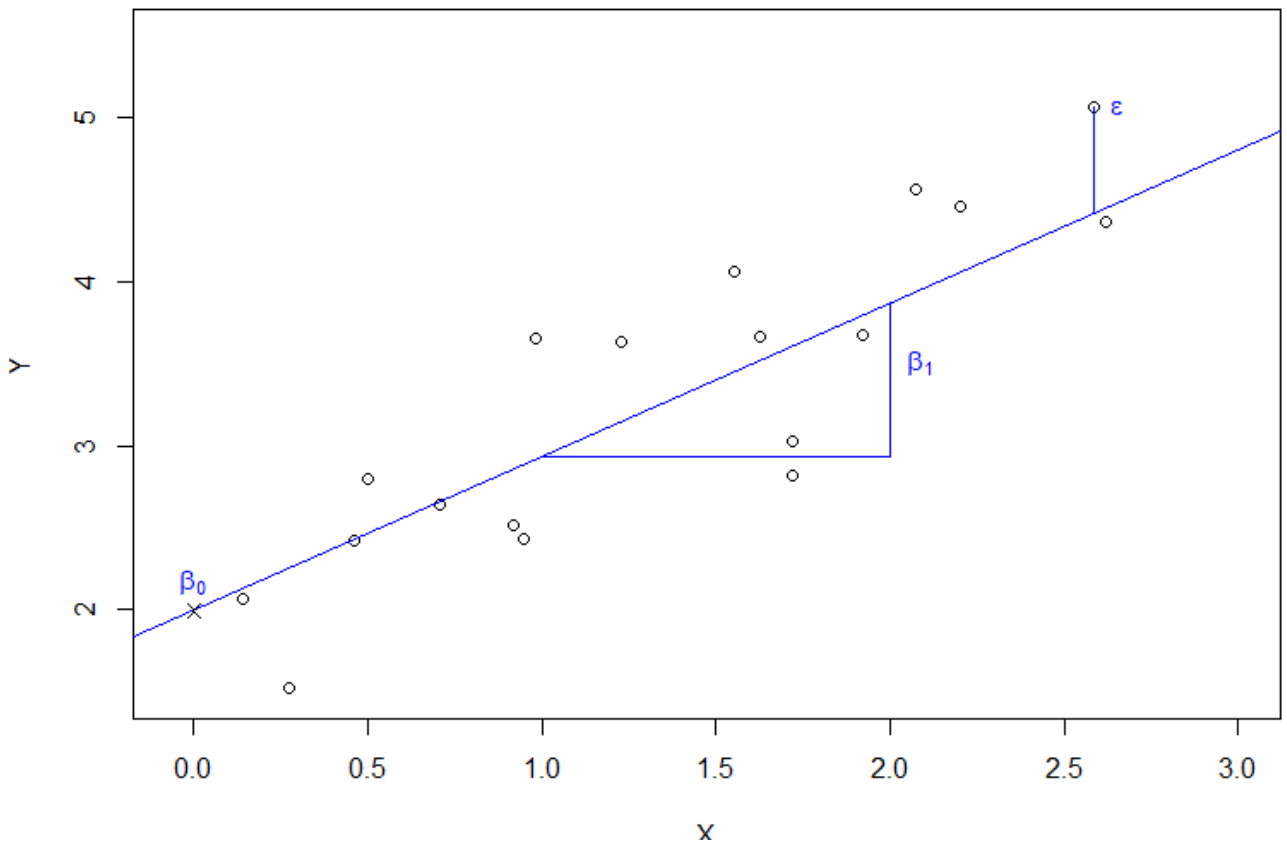
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- **Linearity:** there is a linear relationship between X and y
- **Normality:** the error terms follow a normal distribution
- **Homoscedasticity:** The error variance is constant
- Notice the normality and constant variance assumption concern the error terms.
- These can be thought of as the **true** residual (i.e., difference between observed values and the true fitted line defined by the population parameters)
- These are unknown, so we make do with the **observed** or **fitted** residuals.

The simple linear regression model

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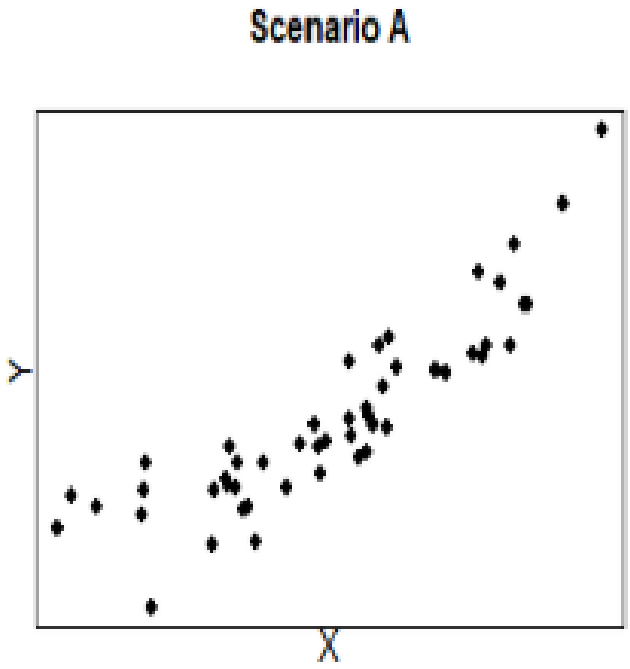
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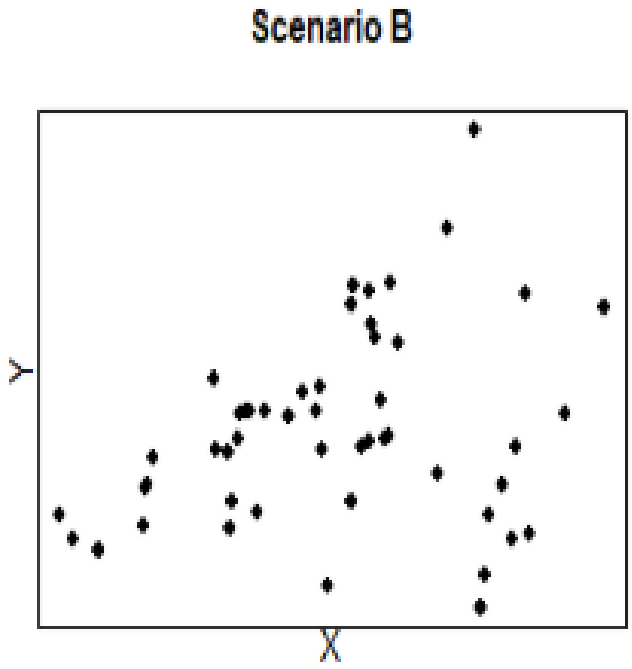
- $\hat{\beta}_0 = -10.75$
- $\hat{\beta}_1 = 0.47$

Scatter plots

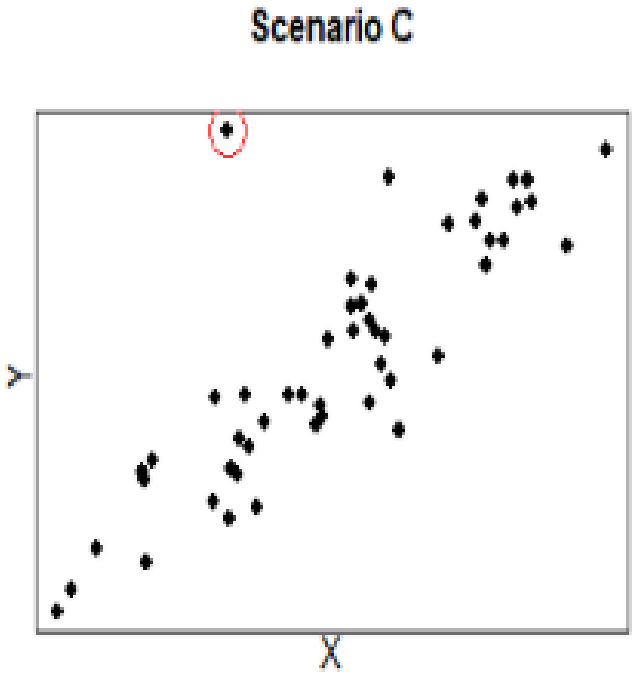
- For simple linear regression, a scatter plot of the outcome against the independent variable can usually make serious violations apparent.
- Such plots are good for identifying non-linearity, heteroscedasticity and outliers.

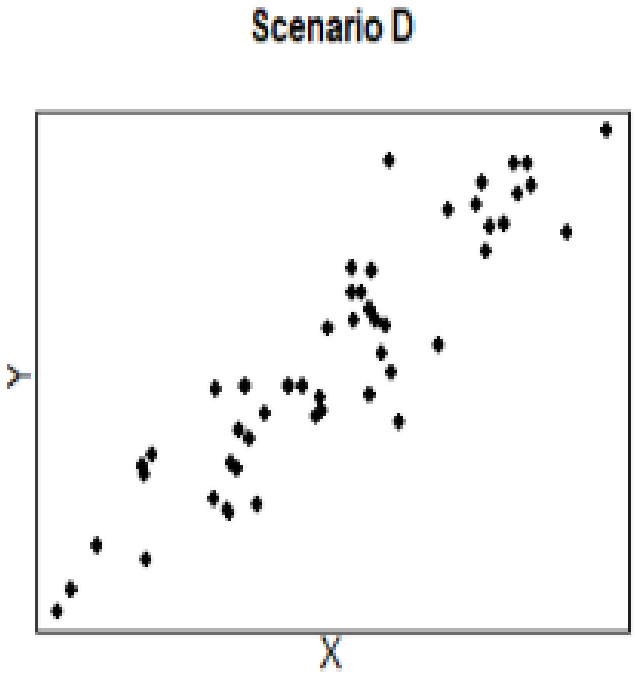
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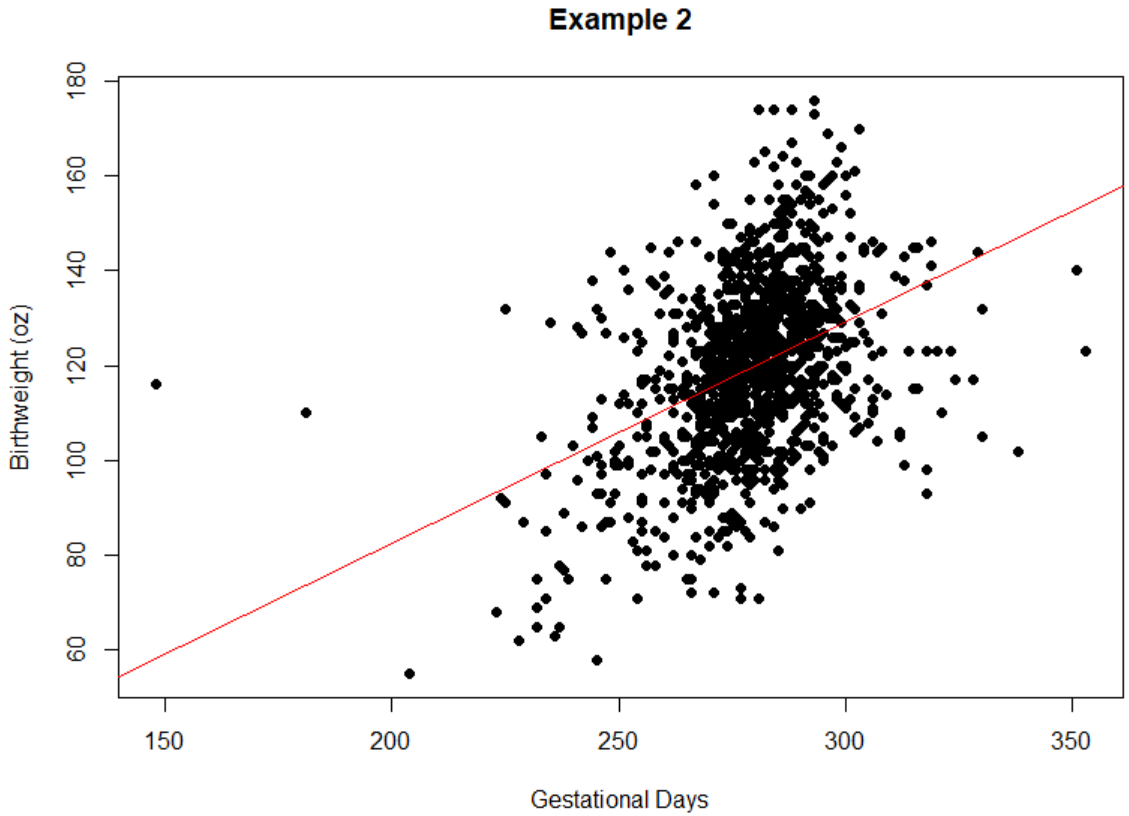


Scatter plots





Example: checking assumptions

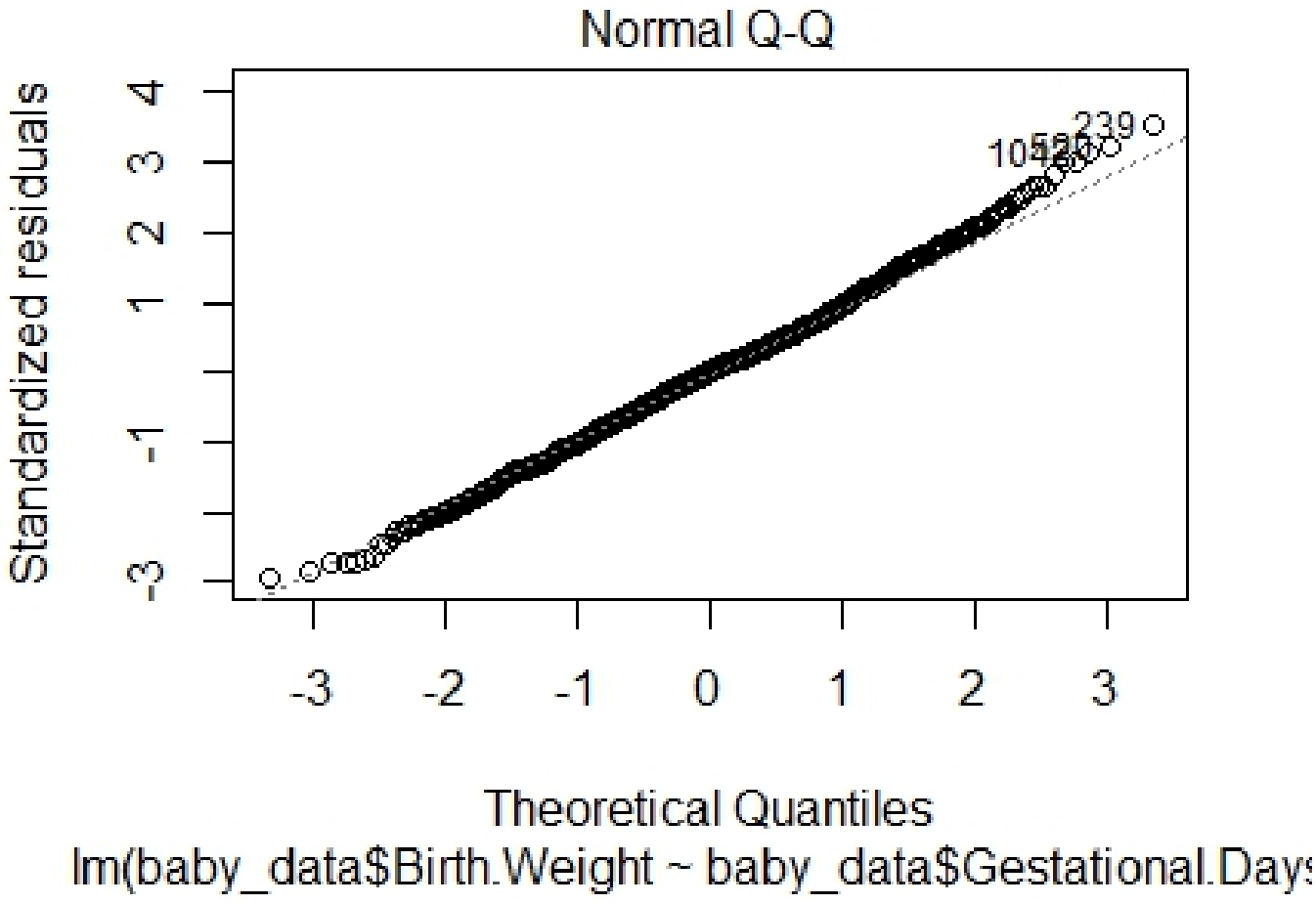


- Variations of linearity and homoscedasticity hold.
- There are potential outliers
- Should consider checking sensitivity to outliers.

Example: checking assumptions

- QQ-plot plots observed values against a standard normal distribution with the same number of points.
- If the data are perfectly normally distributed, the points on a QQ-plot would lie on the line $Y = X$
- Can expect some deviation in the tails

Example: checking assumptions



Recap Diagnostics

The simple linear regression equation:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i \text{ where } \epsilon_i \sim NID(0, \sigma^2)$$

Assumptions:

- Linearity
- Normality
- Homoscedasticity
- Independence

The multivariable linear regression model

- Straightforward extension of the simple linear regression model to situations with multiple covariates.
- Multivariable linear regression equation (assuming p covariates):

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + ... + \beta_p X_{pi} + \epsilon_i \text{ where } \epsilon_i \sim NID(0, \sigma^2)$$

Intercept:

- β_0 is the expectation of Y when all the X_j's are zero.

Slope parameters:

- β_j is the expected change in Y for a unit increase in X_j *when all other covariates are held constant*.

Example: define the model

Let Y_i, X_{1i} and X_{2i} denote the birthweight, length of pregnancy and mother's smoking status for individual i, respectively. We are then interested in the following model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

where $\epsilon_i \sim N(0, \sigma^2)$. $X_{2i} = 0$ for non-smokers and $X_{2i} = 1$ for smokers. The regression parameters are β_0, β_1 and β_2.

Example: R output

```
> multi_model<-lm(data$Birth.Weight~data$Gestational.Days+data$Mat
ernal.Smoker)
> summary(multi_model)

Call:
lm(formula = data$Birth.Weight ~ data$Gestational.Days + data$Mate
rnal.Smoker)

Residuals:
    Min      1Q  Median      3Q     Max
-50.789 -11.035  -0.211  10.053  52.412

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)               -3.18492    8.32945  -0.382    0.702
data$Gestational.Days      0.45117    0.02968  15.200   <2e-16
data$Maternal.SmokerTrue -8.37440    0.97346  -8.603   <2e-16
```

- $\hat{\beta}_0 = -3.18$
- $\hat{\beta}_1 = 0.45$ and $\hat{\beta}_2 = -8.37$

Example: interpreting the parameters

- $\hat{\beta}_0 = -3.185$
- The expected birthweight of a child born after 0 gestational days by a non-smoking mother is -10.75oz
- $\hat{\beta}_1 = 0.45$
- The mean birthweight of a baby is estimated to increase by 0.47oz for each daily increase in the gestational period (among mother's with the same smoking status)
- $\hat{\beta}_2 = -8.37$
- The mean birthweight of a baby from a mother who smokes is expected to by 8.37oz less than the mean birthweight of a baby from a mother who does not smoke (among mother's the same age)

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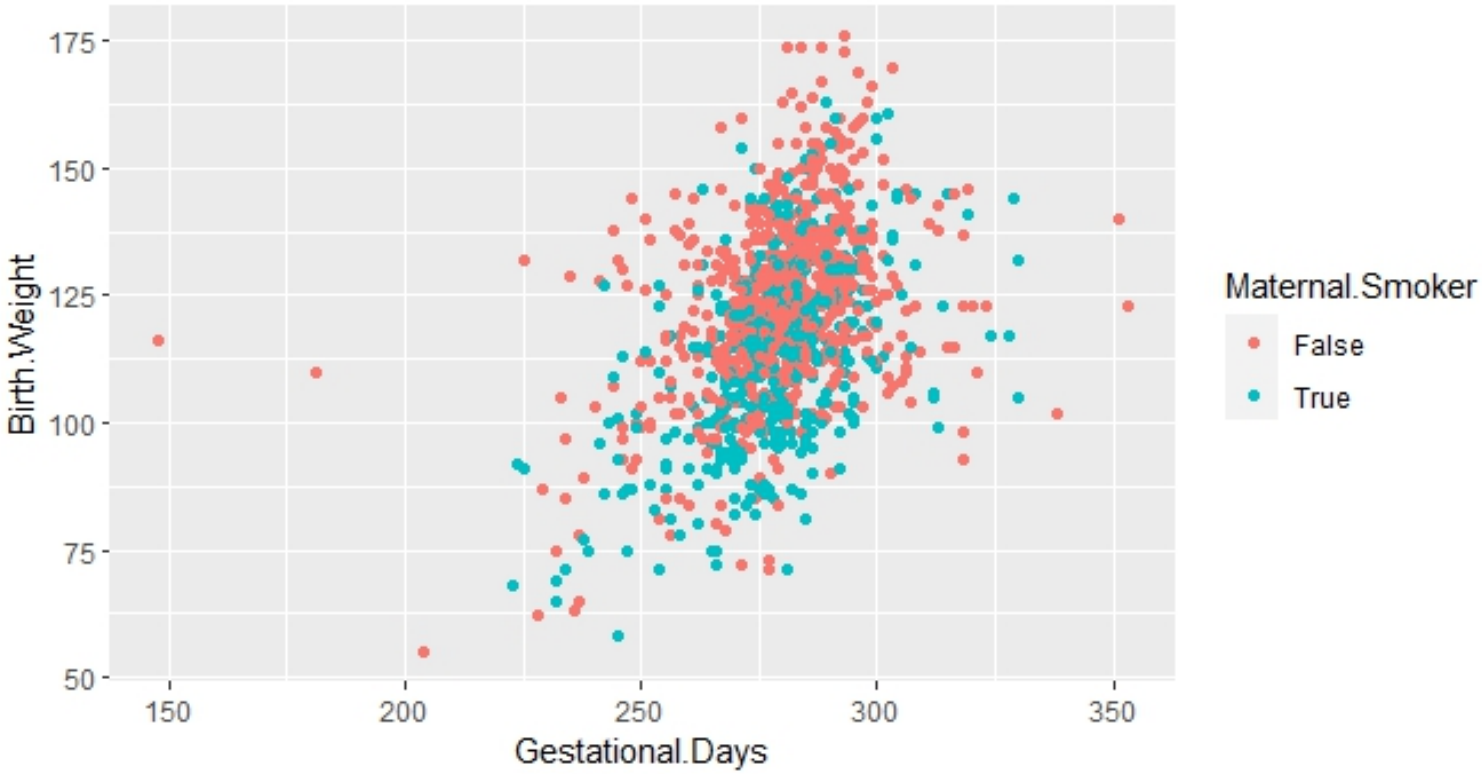
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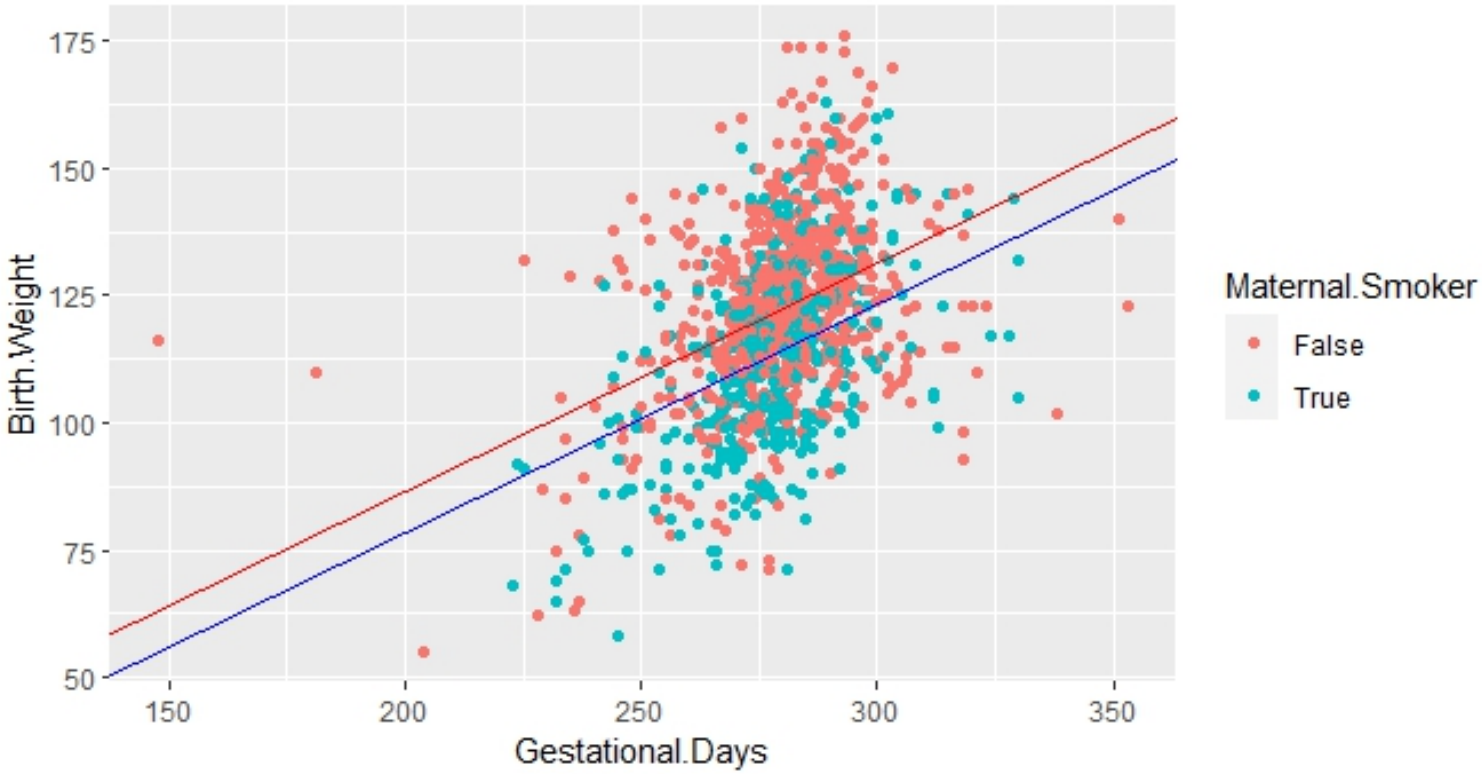
where $\epsilon_i \sim N(0, \sigma^2)$. $X_{2i} = 0$ for non-smokers and $X_{2i} = 1$ for smokers. The regression parameters are β_0, β_1 and β_2.

- $Y_i = \beta_0 + \beta_1 X_{1i}$ for non-smokers
- $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2$ for smokers
- Easier to estimate fitted values

Example: results



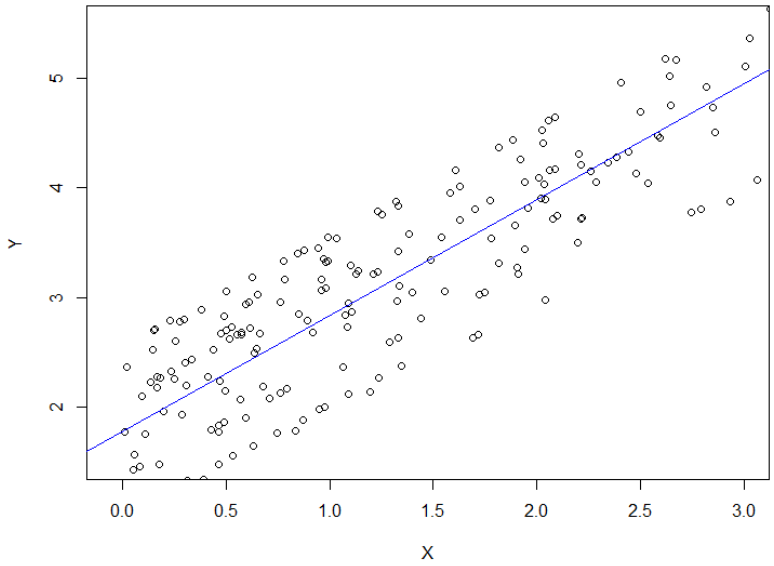
Example: results



Reminder of the assumptions

Four assumptions:

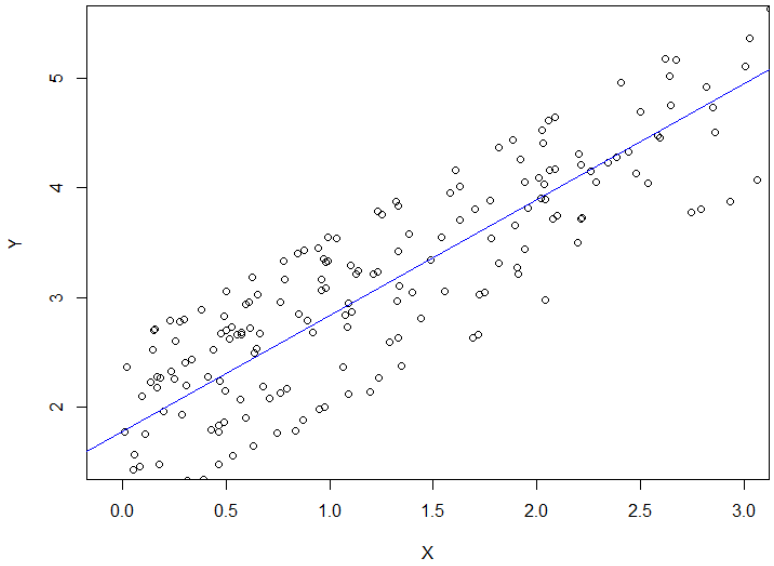
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Four assumptions:

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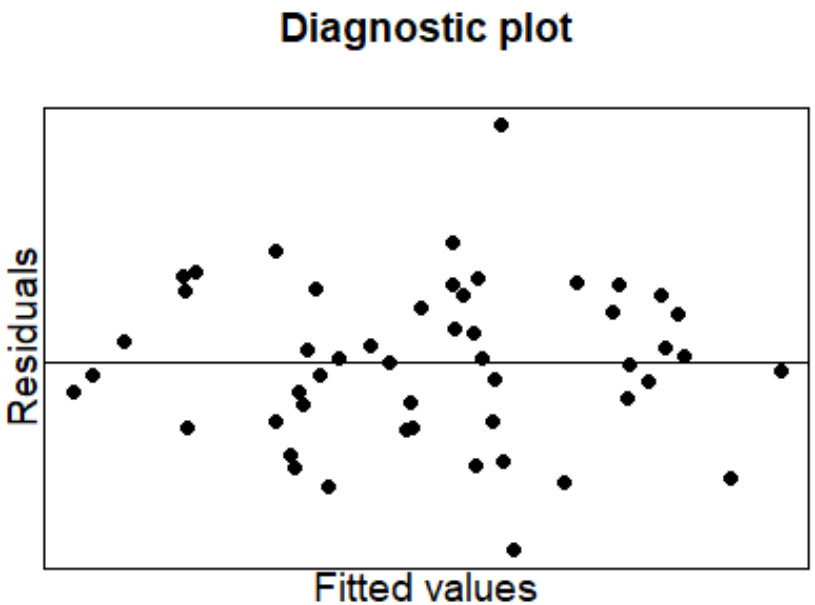
Violations of the independence assumption are often more apparent from the context of a study than from the data itself.

We focus on graphical tools to assess the first three of these assumptions.

Residuals v fitted values

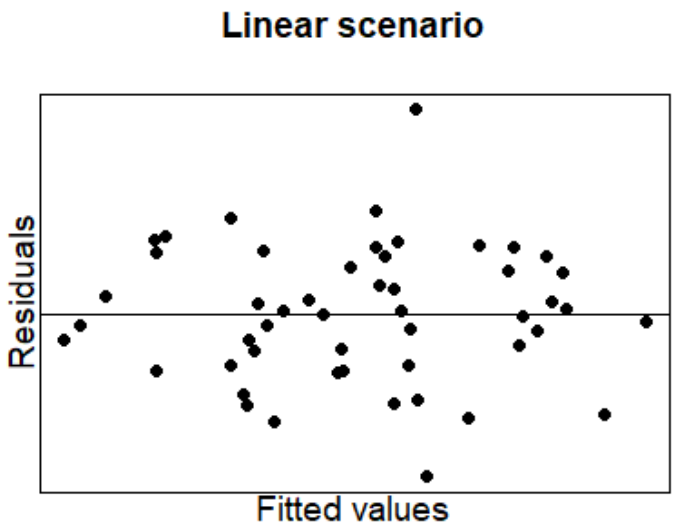
- A plot of the residuals ($\hat{\epsilon}$) against the fitted values ($\hat{y}$) can be used to assess linearity and homoscedasticity.

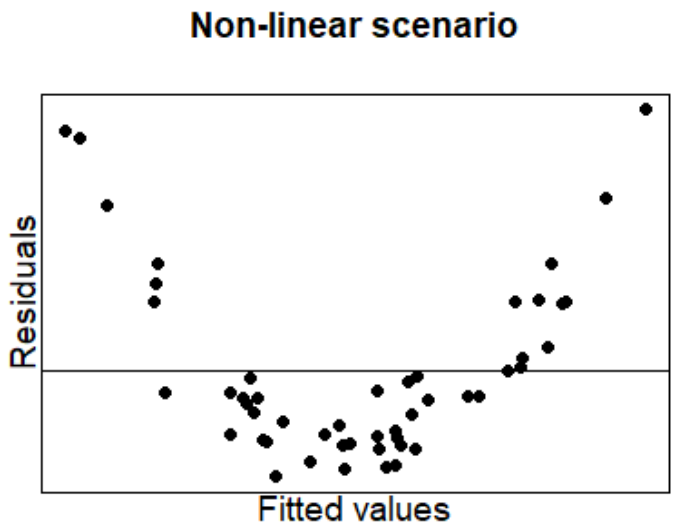
- Residuals should:
  - be equally distributed above and below the zero line
  - have constant variance across the fitted values.



Residuals v fitted values

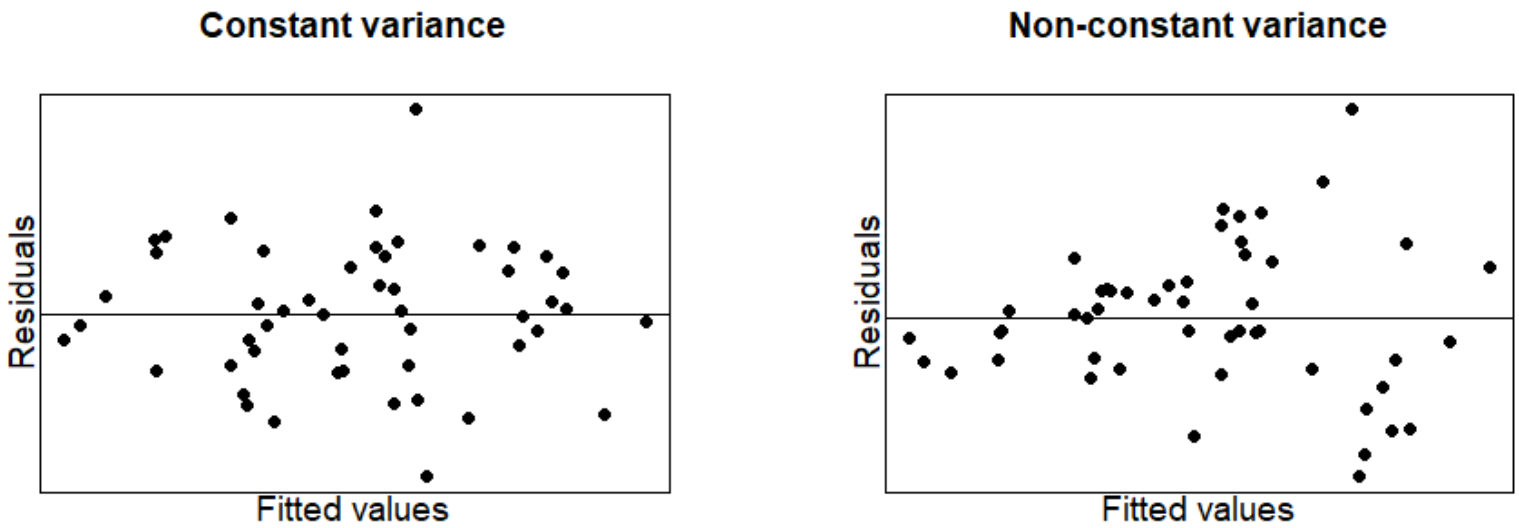
- Unequal distribution of residuals about the zero line indicates non-linearity





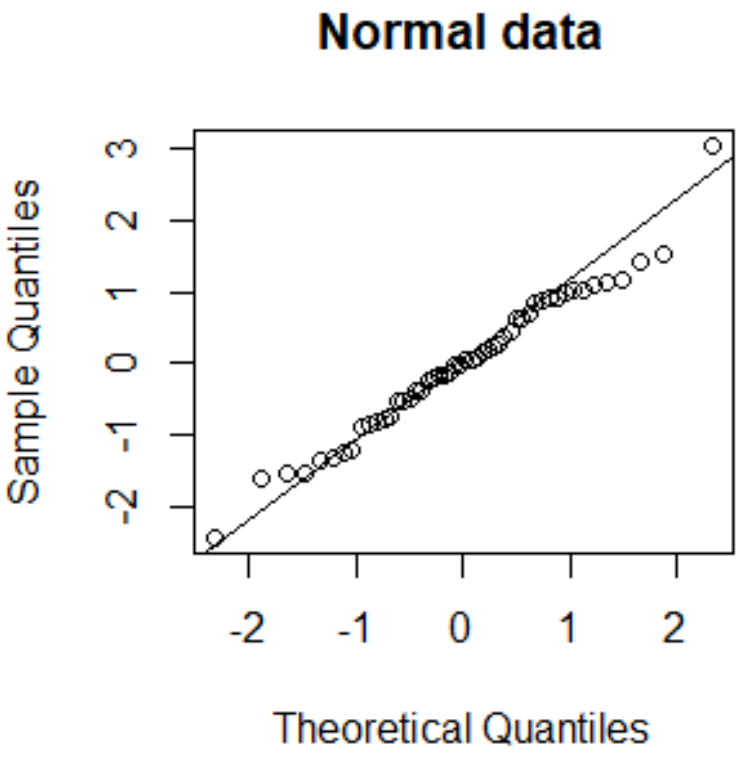
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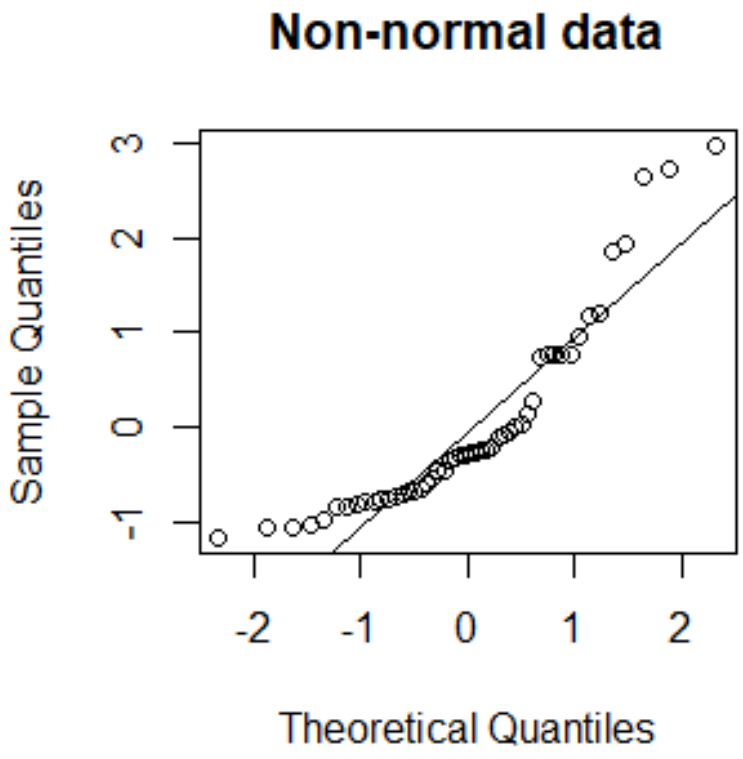
- Non-constant variance across the fitted values indicates heteroskedasticity



QQ-Plots

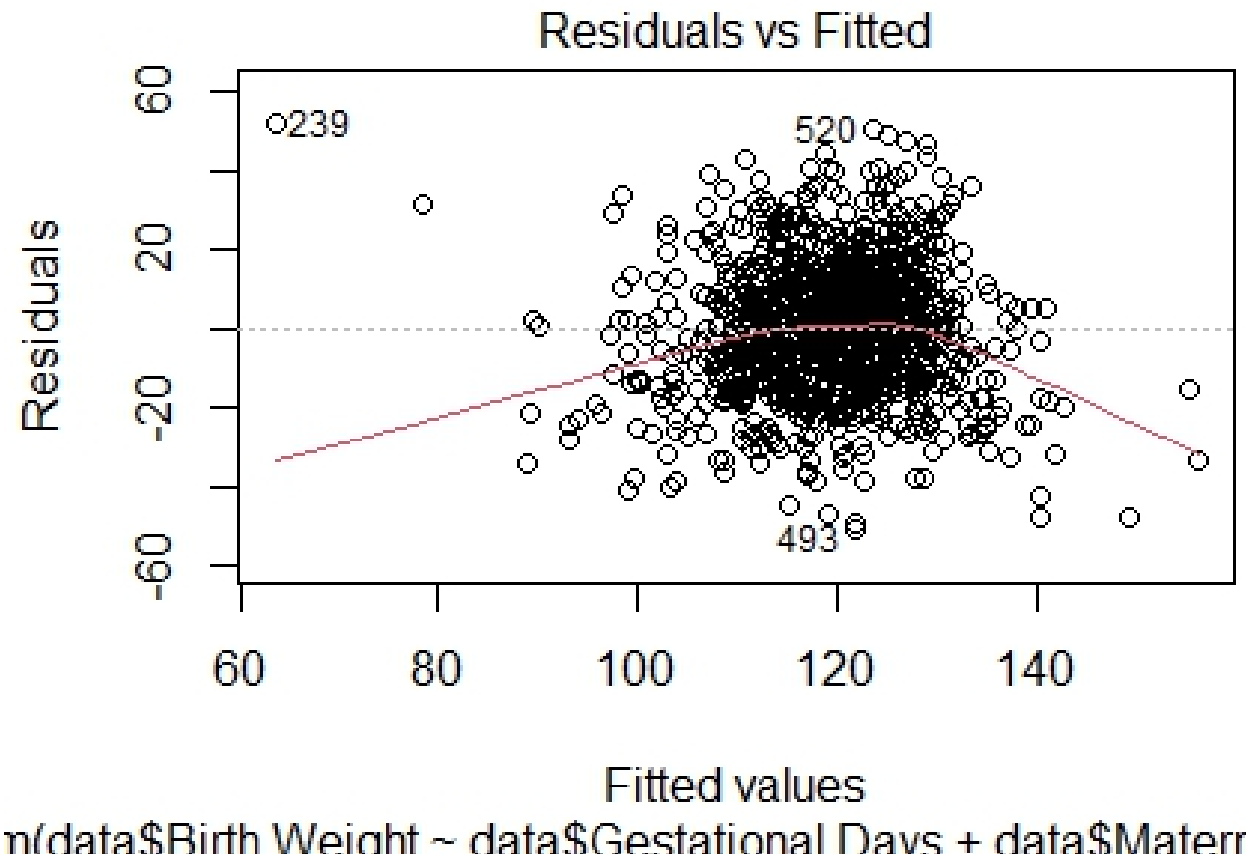
- QQ plots can be used to assess normality
- The data points should lie on a straight line.





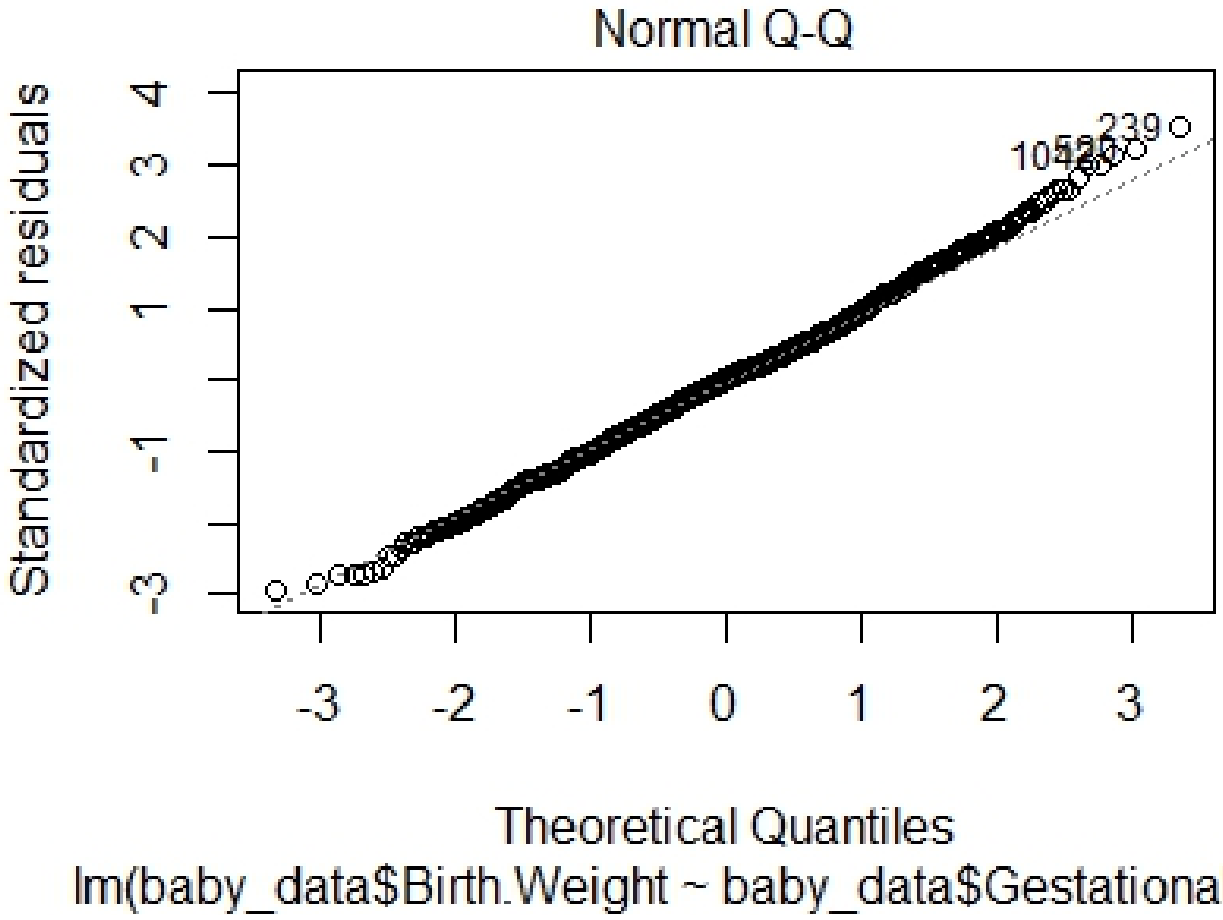
Example: checking assumptions

- Residual plot can be obtained using plot("model name", which=1)



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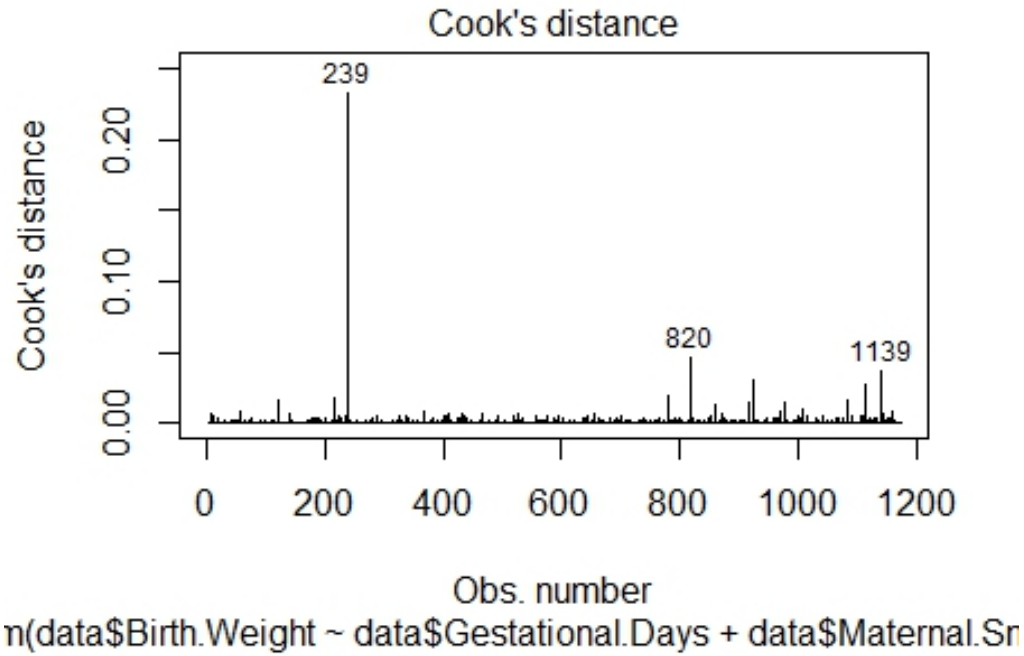
- QQ plot can be obtained using plot("model name", which=2)



Cook's distance

- Cook's distance is a measure of how **influential** individual observations are (use which=4 option for these).

- Very influential observations (relative to the majority) are worth further investigation.



Dealing with violations

- **Transformations** of either the outcome variable and/or one or more of the covariates can sometimes be useful.

- Transformations can be used to deal with violations of normality and homoscedasticity.

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- Advanced methods that relax the linearity assumption (polynomial regression, regression splines)

Dealing with violations

- **Transformations** of either the outcome variable and/or one or more of the covariates can sometimes be useful.
- Transformations can be used to deal with violations of normality and homoscedasticity.
- **Robust methods** may be required (e.g., bootstrapping or robust standard errors)
- Advanced methods that relax the linearity assumption (polynomial regression, regression splines)
- **Sensitivity analyses**

However...

- It is rare for all the assumptions of a statistical procedure to hold exactly.

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- It is rare for all the assumptions of a statistical procedure to hold exactly.
- Broady speaking, the central limit theorem implies that departures from assumptions are less important for large datasets than for small ones.
- Assumption violations are less of a concern when working with big data.

Statistical tests

- There exist a number of statistical tests for normality or the homoscedasticity assumption.
- These tests only have statistical power to detect violations when datasets are large.
- When datasets are large, the central limit theorem means that the consequences of these violations may be practically unimportant.
- For these reasons, the tests are considered by many statisticians to be of limited practical use.

Non-linear terms

Can add higher-order terms to your model, to relax the assumption that Y and X are linear.

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Let Y_i and X_{1i} denote the birthweight and length of pregnancy for individual i, respectively. Then a quadratic model is defined as:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{1i}^2 + \epsilon_i$$

where $\epsilon_i \sim N(0, \sigma^2)$ and β_0, β_1 and β_2 are the regression parameters.

Interaction terms

- Interaction terms are used in situations where the relationship between Y and X differs according to the level of one or more other covariates.

- In linear regression models, the slope of the relationship between Y and X may depend on another factor.

Example: define the model

Let Y_i, X_{1i} and X_{2i} denote the birthweight, length of pregnancy and mother's smoking status for individual i, respectively. Then we can define the following model which includes an interaction term:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{1i} \times X_{2i} + \epsilon_i$$

where $\epsilon_i \sim N(0, \sigma^2)$ and β_0,...,β_3 are the regression parameters.

Example: no interaction term

A model with no interaction term is given by:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i$$

where $\epsilon_i \sim N(0, \sigma^2)$. $X_{2i} = 0$ for non-smokers and $X_{2i} = 1$ for smokers.

Example: no interaction term

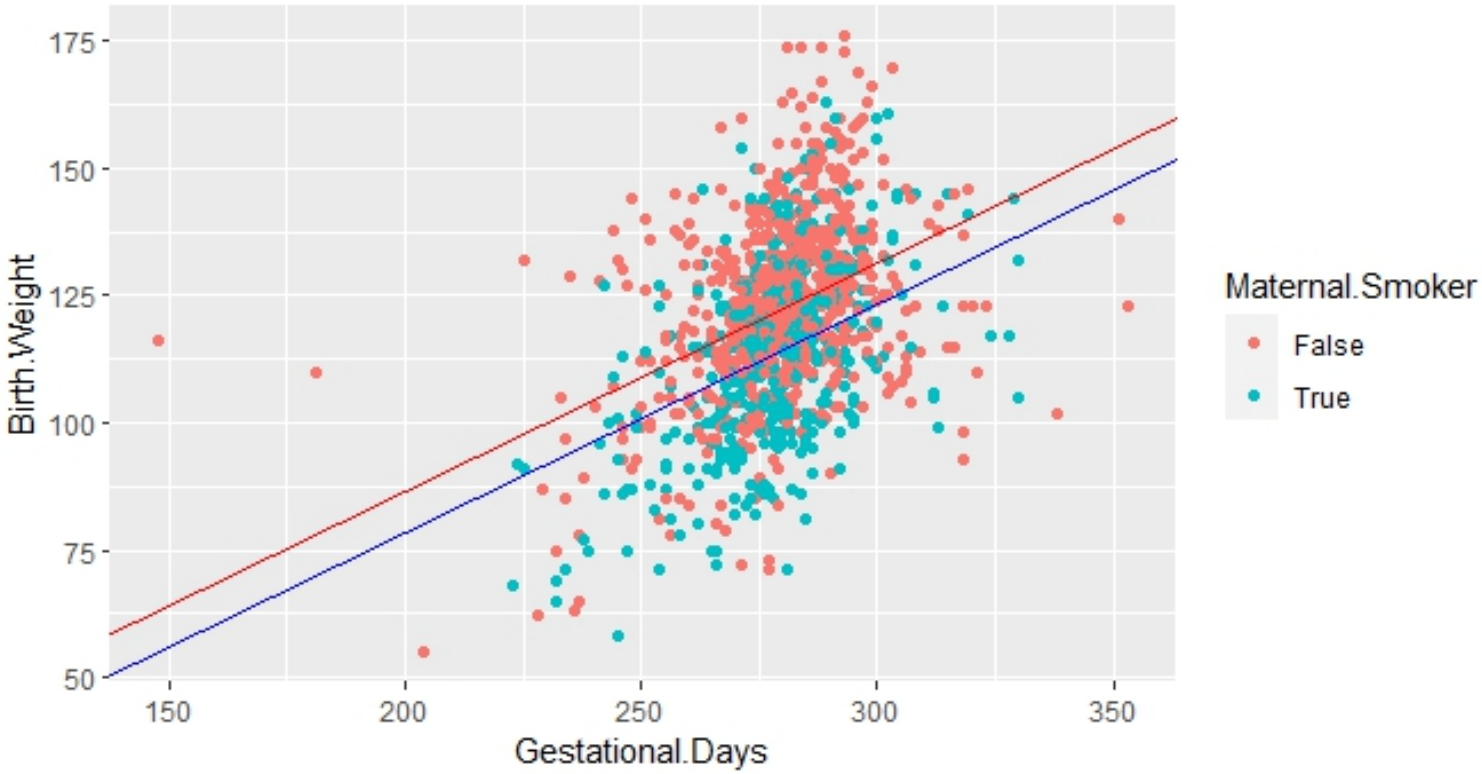
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- $Y_i = \beta_0 + \beta_1 X_{1i}$ for non-smokers
- $Y_i = (\beta_0 + \beta_2) + \beta_1 X_{1i}$ for smokers

Example: fitted values



Example: interaction term

A model including an interaction term is given by:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 (X_{1i} \times X_{2i}) \epsilon_i$$

where $\epsilon_i \sim N(0, \sigma^2)$. $X_{2i} = 0$ for non-smokers and $X_{2i} = 1$ for smokers.

Example: interaction term

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- $Y_i = \beta_0 + \beta_1 X_{1i}$ for non-smokers
- $Y_i = (\beta_0 + \beta_2) + (\beta_1 + \beta_3) X_{1i}$ for smokers

Example: R output

```
> int_model<-lm(data$Birth.Weight~data$Maternal.Smoker*data$Gestational.Days)
> summary(int_model)

Call:
lm(formula = data$Birth.Weight ~ data$Maternal.Smoker * data$Gestational.Days)

Residuals:
    Min      1Q  Median      3Q     Max
-51.023 -11.078  -0.084   9.995  50.499

Coefficients:
                                                Estimate Std. Error t value Pr(>|t|)
(Intercept)                                     19.63964   10.29098   1.908 0.056580
data$Maternal.SmokerTrue                       -72.68713   17.23243  -4.218 2.65e-05
data$Gestational.Days                            0.36962    0.03671  10.069  < 2e-16
data$Maternal.SmokerTrue:data$Gestational.Days   0.23085    0.06176   3.738 0.000194
```

- The above output provides evidence that there is an interaction

Example: fitted values



Collinearity

A potential issue with linear regression is collinearity. This occurs when there is correlation between one of more of the independent covariates. If the degree of correlation is high enough, it can cause the following problems:

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A potential issue with linear regression is collinearity. This occurs when there is correlation between one of more of the independent covariates. If the degree of correlation is high enough, it can cause the following problems:

- Estimated coefficients can swing in either direction.
- Increase variance of estimated coefficients.

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When do we care about collinearity

- Collinearity increases with the degree of correlation, so it is not necessarily a concern if correlation is moderate or weak.
- Including higher-order terms or interaction terms can result in collinearity.
- Collinearity rarely affects the predicted outcomes.
- But it only affects the specific independent variables that are correlated.

What can we do about it

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- Instead of using systolic and diastolic blood pressure as collinear predictors, use diastolic and (systolic-diastolic) blood pressure
- Instead of using height and weight as predictors, using BMI (a combination of height and weight)
- When fitting a quadratic regression model, using X and $(X - \bar{X})^2$, instead of X and X^2.

Summary of lesson

We have recapped some key topics in simple linear regression and multivariable linear regression.

What we have learnt

- How to apply a range of graphical techniques to investigate the assumptions of the linear regression models
- Be able to identify potentially problematic outliers.
- Some techniques for dealing with assumption violations.